AIS – AURELIA INDUSTRIAL SYSTEMS

AIS-100 TECHNICAL SPECIFICATIONS

Aurelia Industrial Systems (AIS)
AIS-100 Technical Specifications

Document ID: AIS-TECH-100
Version: 3.0
Effective Date: January 20, 2026
Category: Technical Documentation
Language: English
Classification: Internal / Customer Reference

---

1. Document Purpose

This document defines the technical specifications and engineering characteristics of the AIS-100 Compact Assembly System.

It is intended for customers, system integrators, maintenance personnel, technical support engineers, and AIS engineering teams.

The document provides technical reference information for installation, operation, integration, maintenance, and troubleshooting.

This document does not replace the AIS-100 User Manual or applicable safety documentation.

---

2. System Overview

The AIS-100 is a compact automated assembly system designed for continuous industrial production environments.

The system integrates:

- automated material handling;
- assembly stations;

- programmable control logic;
- operator HMI;
- safety sensors;
- emergency stop circuits;
- industrial communication interfaces;
- production monitoring functions.

The AIS-100 is designed for applications where a compact machine footprint and moderate production throughput are required.

2.1 Product Identification

Parameter	Specification
Product	AIS-100
Machine type	Compact Assembly System
Nominal maximum capacity	40 units/min
Control system	AIS Control PLC-100
HMI	AIS Industrial HMI
Primary communication	Industrial Ethernet
I/O	Digital I/O
Safety architecture	Dual-channel safety circuits
Preventive maintenance interval	Every 1,500 operating hours

---

3. Mechanical Specifications

3.1 Dimensions

The standard AIS-100 configuration has the following approximate dimensions:

Parameter	Value

Length	2,200 mm
Width	1,400 mm
Height	2,100 mm
Approximate weight	1,850 kg

Actual dimensions may vary depending on the selected machine configuration and optional equipment.

Adequate clearance must be provided around the machine for operation, inspection, maintenance, and emergency access.

3.2 Mechanical Configuration

The standard machine consists of:

- main structural frame;
- assembly area;
- material handling system;
- actuator system;
- electrical cabinet;
- operator interface;
- safety guarding.

Optional mechanical configurations may be available depending on the customer application.

---

4. Electrical Specifications

The standard electrical configuration is:

Parameter	Specification
Supply voltage	400 VAC
Supply type	Three-phase

Frequency	50/60 Hz
Control voltage	24 VDC
Main control system	AIS Control PLC-100

The installation must provide a suitable electrical supply according to the machine configuration and applicable local requirements.

Electrical installation and intervention must be performed by qualified personnel.

The machine must be properly grounded before operation.

---

5. Performance and Capacity

The AIS-100 has a nominal maximum production capacity of:

40 units per minute

Actual production throughput depends on:

- product characteristics;
- material availability;
- machine configuration;
- cycle time;
- operator intervention;
- upstream and downstream equipment;
- production recipe;
- environmental conditions.

The nominal capacity should not be interpreted as a guaranteed production rate for every application.

---

6. Environmental Requirements

The standard operating environment is:

Parameter	Requirement
Operating temperature	5–35 °C
Relative humidity	Up to 80%, non-condensing
Installation environment	Indoor industrial environment
Altitude	Up to 1,000 m without special configuration

The machine should not be installed in environments containing excessive dust, corrosive substances, or conditions outside the specified operating range unless the configuration has been specifically approved by AIS Engineering.

---

7. Control System

The AIS-100 uses the **AIS Control PLC-100** as its primary programmable control system.

The PLC manages:

- machine sequencing;
- actuator control;
- sensor monitoring;
- production cycle logic;
- machine states;
- alarm handling;
- communication with the HMI;
- safety-related machine status signals.

The control software configuration is machine-specific and may depend on the customer production configuration.

Unauthorized modifications to PLC programs are prohibited.

---

8. Safety System

The AIS-100 incorporates a safety architecture designed to prevent hazardous machine operation when defined safety conditions are not satisfied.

Safety components include:

- dual-channel emergency stop circuits;
- guard interlocks;
- safety sensors;
- monitored machine states.

Activation of an emergency stop must place the machine in a safe state.

Safety devices must not be bypassed, disabled, or modified without an approved engineering procedure.

Any safety-related intervention must be performed by qualified personnel.

---

9. Communication Interfaces

The standard AIS-100 communication interfaces include:

- Industrial Ethernet;
- digital I/O.

Industrial Ethernet may be used for communication with compatible industrial systems.

Digital I/O interfaces are available for machine-level integration with external equipment where supported by the configured machine architecture.

Communication configuration is application-dependent.

---

10. HMI and Diagnostics

The AIS Industrial HMI provides operators with access to:

- machine status;
- production information;
- operating states;
- alarms;

- basic diagnostic information;
- maintenance-related information.

Alarm information should be reviewed before attempting corrective action.

The HMI does not replace qualified technical diagnosis for electrical, mechanical, safety, or control-system failures.

---

11. Utilities and Auxiliary Requirements

The standard AIS-100 configuration requires electrical power and may require auxiliary utilities depending on the machine configuration.

Where pneumatic equipment is installed, the required compressed-air specification must be verified against the machine configuration supplied to the customer.

Project-specific utility requirements are defined during machine configuration and installation planning.

Customers must ensure that supplied utilities meet the requirements specified in the approved installation documentation.

---

12. Maintenance Requirements

Preventive maintenance for the AIS-100 is scheduled every:

1,500 operating hours

Typical preventive maintenance activities include:

- inspection of mechanical components;
- inspection of safety devices;
- inspection of sensors;
- inspection of electrical connections;
- cleaning;
- inspection of actuators;
- verification of machine operation.

Maintenance activities must follow the applicable AIS maintenance documentation.

Operating-hour information shown by the machine must be considered when determining the maintenance status.

---

13. Installation Considerations

Installation must provide:

- suitable floor support;
- sufficient service clearance;
- appropriate electrical supply;
- appropriate grounding;
- suitable environmental conditions;
- access for maintenance;
- safe operator access;
- suitable upstream and downstream interfaces.

Final installation requirements depend on the machine configuration and customer production environment.

The machine must be commissioned according to the applicable AIS installation and commissioning procedure.

---

14. Configuration and Variants

The AIS-100 may be configured according to customer production requirements.

Possible configuration differences include:

- assembly tooling;
- material handling configuration;
- sensor configuration;
- HMI configuration;
- communication configuration;
- optional production equipment.

Configuration-specific information takes precedence over generic specifications where the two differ.

---

15. Technical Support Data

When contacting AIS Technical Support, provide:

- machine model;
- machine serial number;
- current operating hours;
- HMI alarm or error code;
- description of the issue;
- production impact;
- recent maintenance activity;
- relevant diagnostic information.

For recurring or intermittent problems, provide the approximate time and operating conditions under which the issue occurred.

---

16. Document Control

Version	Date	Description
2.0	2025-01-15	Technical specification revision
2.5	2025-07-10	Updated communication and safety information
3.0	2026-01-20	Current technical specification

Current approved version: 3.0